

CONTACT TOPOLOGY WORLD MODELS: ICLR MAIN-CONFERENCE GATE ARCHIVE

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ABSTRACT

This document is not an ICLR main-conference submission. It is the v3 main-conference gate for a generated robotics draft about contact-rich robot world models. The original research bet was: Predict topology changes in contact graphs rather than only state trajectories. After a stricter hostile review, the honest terminal decision is **KILL/ARCHIVE**. Submission-hardening version: v3.

1 GATE DECISION

The v2 paper was workshop-only. Under the active ICLR-main standard, that is insufficient. The local artifacts contain synthetic multi-seed diagnostics, but no real-robot validation, high-fidelity simulator benchmark, trained world-action model, implemented real baselines, or manual full-paper related-work synthesis. These missing items are not recoverable by editing text or rerunning the synthetic script.

2 FATAL ICLR-MAIN BLOCKERS

- Evidence is synthetic and generated from a shared batch template.
- Baselines are diagnostic probability models, not implemented competing robot systems.
- Ablations do not correspond to a trained model checkpoint.
- No hardware, high-fidelity benchmark, qualitative rollouts, or failure videos are available.
- Related work is based on pool slices and hostile metadata, not a manual full-paper synthesis.
- Claims would be misleading if framed as ICLR-main empirical robotics evidence.

3 HOSTILE PRIOR-WORK PRESSURE

The closest local hostile records include:

- OmniVTA: Visuo-Tactile World Modeling for Contact-Rich Robotic Manipulation (2026)
- Do World Action Models Generalize Better than VLAs? A Robustness Study (2026)
- PolyTouch: A Robust Multi-Modal Tactile Sensor for Contact-rich Manipulation Using Tactile-Diffusion Policies (2025)
- Contact-implicit Model Predictive Control: Controlling diverse quadruped motions without pre-planned contact modes or trajectories (2025)
- Force Policy: Learning Hybrid Force-Position Control Policy under Interaction Frame for Contact-Rich Manipulation (2026)

These works make generic world models, uncertainty, model-based planning, and robot policy robustness crowded. Without real empirical evidence, the remaining novelty boundary is too weak for ICLR main.

4 WHAT REMAINS USEFUL

The archive is still useful as an idea seed and reproducibility scaffold. The synthetic code, stress-test CSVs, reviewer attacks, and novelty-boundary docs can inform a future project. The correct next step is not more paper polishing; it is new evidence: implemented models, real baselines, manual related work, and robot or high-fidelity simulator validation.

5 TERMINAL ACTION

Decision: **KILL/ARCHIVE for ICLR main**. Do not submit this paper to ICLR main in the current form. Revival requires a substantially new empirical project.